

SYED MUHAMMAD HAMZAH RIZVI

AI Reliability & Test Automation Engineer

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PROFESSIONAL SUMMARY

Engineer with 3+ years building and validating production software, specialising in test design, regression frameworks and evaluation pipelines for AI/LLM systems. Built a 140+ test pure-stdlib regression suite covering each diagnosed bug by name for a production RAG pipeline, plus a 34-case evaluation harness gating every change — enabling fast, evidence-driven iteration without running model inference on each run. Diagnosed a silent correctness failure in that system's hallucination gate, where the NLI verifier was reading the classifier's neutral class instead of entailment, and took a true/false claim battery from three wrong verdicts to 8/8. Comfortable across Python services, REST APIs and CI/CD, with strong debugging instincts in distributed, non-deterministic systems.

CORE SKILLS

- **Testing & Automation:** Test-driven development, regression suite design, unit/integration/API/end-to-end testing, mocks and fixtures, CI/CD pipelines, debugging distributed and non-deterministic systems.
- **AI/LLM Systems:** RAG pipelines, hybrid retrieval, reranking, grounding/hallucination checks, prompt and conversational-flow evaluation, model output validation.
- **Engineering:** Python, C++, FastAPI, REST API design and debugging, webhooks, authentication flows, JSON, Git, production debugging across APIs, devices and networks.
- **Collaboration:** Working cross-functionally with engineering and product teams, technical documentation, translating diagnosed failure modes into reusable regression coverage.

PROFESSIONAL EXPERIENCE

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| Technical Software Consultant — Innovative Technology, Manchester, UK | Mar 2023 – Present |
| <ul style="list-style-type: none">• Own customer deployments of AI-powered biometric and identity-verification products end to end, diagnosing production issues spanning APIs, devices, mobile apps, firmware and networks.• Design and build REST API integrations, webhooks and authentication flows, then validate behaviour through structured, reproducible test cases before rollout to production.• Trace intermittent and non-deterministic production failures back to root cause across distributed systems, feeding fixes and regression coverage back into product and firmware.• Use AI coding assistants daily to prototype integrations and accelerate debugging, validating output against reproduced behaviour rather than accepting it blindly.• Create reusable technical assets — integration guides, API documentation and test scenarios — that make each deployment faster and safer than the last.• Delivered rollouts across retail, gaming and compliance-focused environments spanning thousands of operational endpoints. | |
| Automation Engineer — Tripack Films Ltd., Pakistan | Mar 2020 – Sep 2021 |
| <ul style="list-style-type: none">• Built and maintained industrial automation workflows, integrating control systems and sensors; diagnosed hardware and software issues in real-time operational environments. | |
| Software Consultant — Evolligence, Pakistan | Dec 2019 – Mar 2020 |
| <ul style="list-style-type: none">• Built and supported backend application logic using structured APIs and data-driven workflows, collaborating with development teams on delivery. | |

SELECTED ENGINEERING PROJECTS

GroundedOps — Production RAG Support Assistant — *Python, FastAPI, React, ChromaDB, Ollama, Docker, DeepSeek/OpenAI/Claude APIs*

- Deployed to production and now serving 300+ customer accounts weekly through a public embeddable widget and admin console; associated with a 40% reduction in first-line ticket volume and a 55% reduction in resolution time per internal Jira metrics since release (tracked alongside other concurrent initiatives, so not isolated to this tool alone).
- Built a retrieval-augmented QA system with hybrid retrieval (BM25 + dense embeddings via reciprocal rank fusion), sigmoid-calibrated cross-encoder reranking and NLI-based grounding checks to reduce hallucination from local LLMs.

- Diagnosed a query-routing bug by reproducing it in code, tracing it to an unscoped conversation-memory store and a weak follow-up heuristic; rebuilt the path with an LLM Rewrite-Retrieve-Read flow and per-session memory.
- Wrote a pure-stdlib regression suite of 140+ tests covering each diagnosed bug by name, plus an evaluation harness gating every change against 34 real test cases — including retrieval ranking, grounding failures and memory-scoping edge cases — enabling fast iteration without running model inference on every change.

Age Estimation & Demographic Analysis — *TensorFlow/Keras, EfficientNetB0, VGG16*

- Built a hierarchical multi-task EfficientNetB0 pipeline (gender, coarse age, fine age and continuous age regression) on 274k+ facial images from WIKI/IMDB, fine-tuned on UTKFace, with GPU-accelerated training and evaluation tooling; achieved 87.5% gender accuracy, 88.4% coarse-age accuracy, 92.6% near-group (± 1) accuracy and 8.52-year MAE.

EDUCATION

MSc Robotics and Automation (Minor in AI) — University of Salford, UK 2021 – 2022

Graduated with Distinction. Dissertation: facial age and gender estimation using CNNs.

BEng Electrical Engineering — NUST, Pakistan 2015 – 2019

Focus: computer vision, machine learning and robotics.